

Normal Stress - Strain

課本

<https://www.mechref.org/sol/>

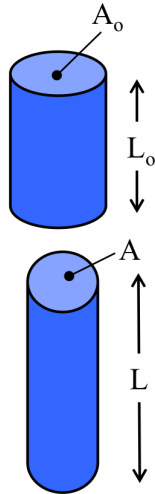
- Stress: force per unit area

$$\sigma = \frac{\text{Force}}{\text{Area}}$$

$$\text{Engineering: } \sigma = \frac{F}{A_0}$$

$$\text{True: } \sigma_T = \frac{F}{A}$$

- Traditional unit: MPa
- A_0 is original area
- A is instantaneous area



(Normal)

- Strain: "relative" change in length

$$\varepsilon = \frac{\Delta \text{Length}}{\text{Length}}$$

$$\text{Engineering: } \varepsilon = \frac{L - L_0}{L_0} = \frac{\Delta L}{L_0}$$

$$\text{True: } \varepsilon_T = \ln\left(\frac{L}{L_0}\right)$$

$$\text{From } d\varepsilon_T = dL/L$$

- Dimensionless quantity
- L_0 is original length ("gage length")
- L is instantaneous length

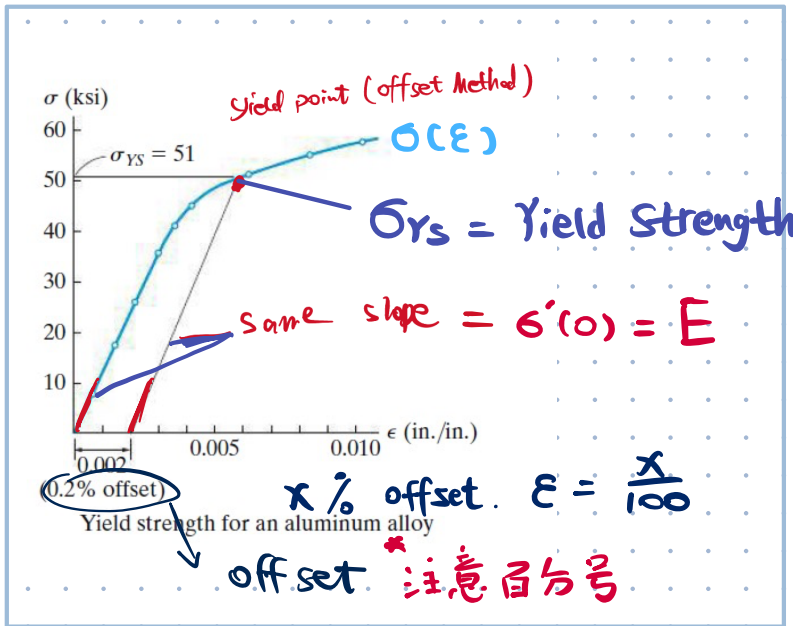
F: 軸向力
 A_0 : 初始面積
 ΔL : 變化長度
 L_0 : 初始長度

Ductile Material Offset Method

—then true and engineering stress and strain are related according to

$$\sigma_T = \sigma(1 + \varepsilon)$$

$$\varepsilon_T = \ln(1 + \varepsilon)$$

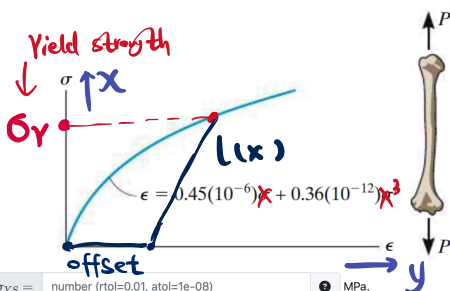


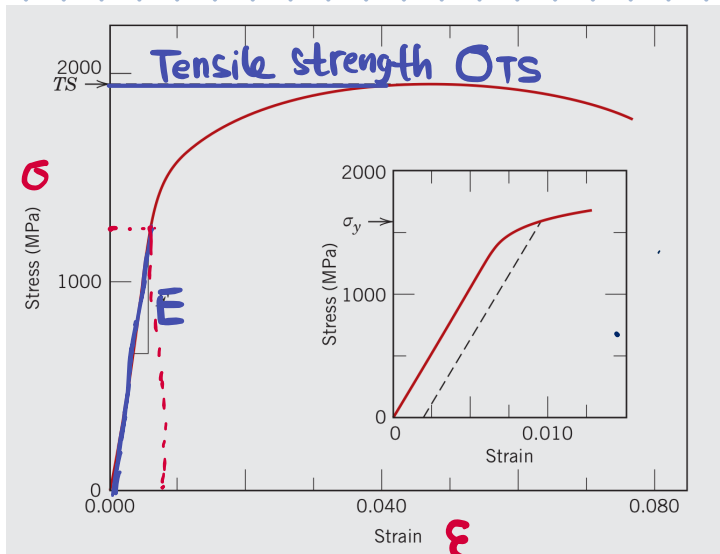
Example Given: $\varepsilon(0)$, $\varepsilon_{\text{off}} = 0.003$ Find: σ_Y

Soln: $x = \sigma$, $y = \varepsilon$
 $\varepsilon_{\text{off}} = 0.003 = b$, $\frac{1}{E} = k = y'(0)$
 $l(x) = kx + b$
 Solve: $y(x) = l(x)$
 $\Rightarrow x = \sigma_Y$

HW3.3. Yield Strength of Bone

The stress-strain diagram for a bone is shown, and can be described by the equation $\varepsilon = 0.45(10^{-6})\sigma + 0.36(10^{-12})\sigma^3$, where σ is in kPa. Determine the yield strength assuming a 0.3% offset.





$$\text{Tensile Strength} = \sigma_{\max}$$

E [Pascal] = Modulus of Elasticity = Stiffness = Young's Modulus

$$\sigma = E \epsilon$$

Poisson's Ratio ν (nu) [dimensionless]

$$\nu = - \frac{\epsilon_{\text{long}}}{\epsilon_{\text{lat}}}$$

* 注意负号

* $0 < \nu \leq 0.5$ (rubber: Const. Volume)

Axial Strain (elongated), applied force

$$\epsilon_{\text{long}} = \frac{\Delta L}{L_0}$$

Radial Strain, $\perp$ applied force

$$\epsilon_{\text{lat}} = \frac{\Delta D}{D_0} \quad (= \frac{\Delta R}{R_0})$$

* Sign Convention

伸长 (elongate) ΔL (ΔD) > 0

压缩

ΔL (ΔD) < 0

ϵ_{long} 与 ϵ_{lat} 符号

Energy

* Not provided in Eqn Sheet

- Strain Density Energy U [Pascal]

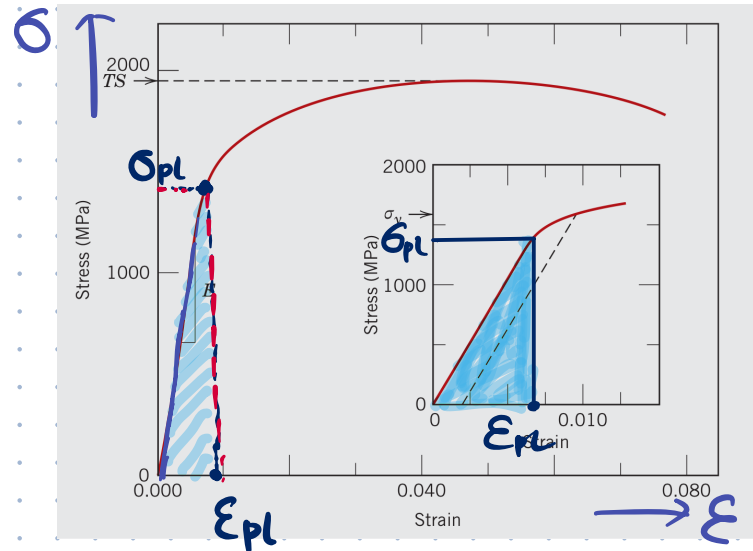
$$U = \int_0^{\epsilon} \sigma(\epsilon) d\epsilon$$

- Modulus of Resilience U_r [Pascal]

Maximum Energy Stored with no harm to material.

$$U_r = \frac{1}{2} \sigma_{PL} \epsilon_{PL}$$

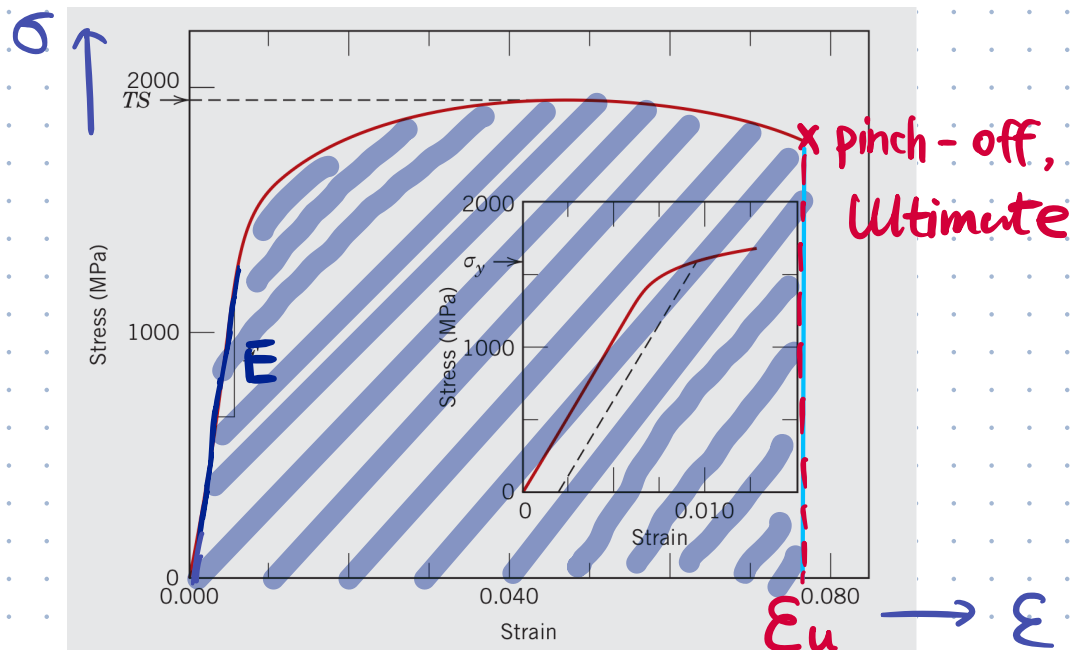
$$= \frac{1}{2} \sigma_{PL}^2 / E$$



- Modulus of Toughness U_t [Pascal]

Maximum Energy Stored before the material fracture

$$U_t = \int_0^{\epsilon_u} \sigma(\epsilon) d\epsilon$$

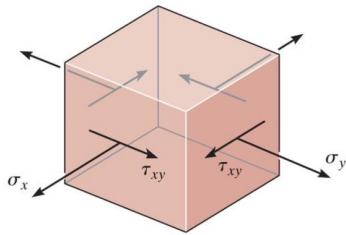
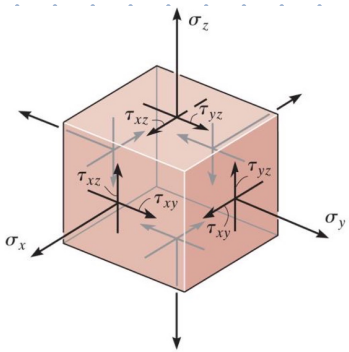


Stress Directions

Normal Stress $\sigma_x, \sigma_y, \sigma_z$ 方向: 垂直面向外

Shear Stress τ_{xy} — Force Direction $\oplus$ x, y 同号
Face Normal (向外) $\ominus$ x, y 异号

$$\tau_{yx} = \tau_{xy}, \text{下标反-反值不变}$$



Factor of Safety Units
见课本 - stress

Shearing Stress - strain

τ (tau) Shearing Stress [Pascal/Psi]

γ (gamma) Shearing strain [dimensionless]

G Shear modulus of Elasticity / Modulus of Rigidity [Pascal/Psi]

* Shear Stress

见课本 - stress

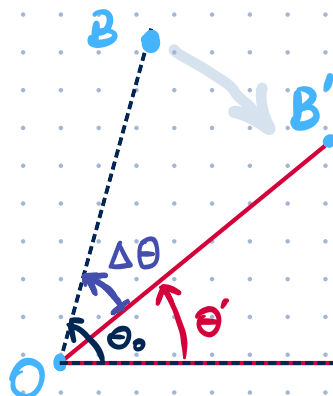
$$\tau = G \gamma$$

$$E = 2(1 + \nu) G$$

Shear Strain

γ (gamma) [dimensionless]

$$\gamma = \tan(\Delta\theta)$$



θ_0 初始 $\angle AOB$
 θ' 变形 $\angle AOB'$
 $\Delta\theta$ 变化角度 $\angle BOB'$